DSAA3073: Theories in Data Science

Tutorial 7A: Confidence at a Common Scale

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-31

Explorative projection

LESSON BACKBONE

Zero training error records only signs. Which scale-invariant quantity keeps the confidence information, and under exactly what class and sample conditions can its empirical distribution control population error?

AdaBoost ends with a real score before it takes a sign. Two ensembles can therefore classify every training example correctly while placing those examples very differently relative to zero. Raw score magnitude is not yet a fair comparison, however, because multiplying every coefficient by the same positive constant changes the score without changing the classifier.

We will first remove that arbitrary scale and keep careful track of what space any geometric distance lives in. We will then turn the normalized training values into a cumulative empirical curve. Only after that evidence is visible will we state a population theorem, with its class, threshold, sample, probability, and complexity conditions attached.

The three core blocks take **50, 40, and 60 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-50	Separate signed score, normalized confidence, and output-space distance	50 min core
50-65	Scheduled break	15 min
65-105	Build and audit an empirical cumulative margin distribution	40 min core
105-165	Assemble a fixed-threshold population margin certificate	60 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

Put every ensemble on a common scale

Chapter 6 made exactly four items available at this boundary, in order. The third item is a verified pass-through from Chapter 2; it was not introduced by AdaBoost.

No.	Available item	Use in this Tutorial
1	AdaBoost	Reconstruct its coefficients, real score, and final sign.
2	AdaBoost training-error product bound	Recall what a small empirical sign-error certificate does and does not report.
3	VC dimension	Supply the capacity of the binary base class in a population bound.
4	three complementary AdaBoost views	Keep algorithmic, optimization, and confidence claims logically separate.

IN-CLASS CHECK

Prompt: Reconstruct the four items in order. Which one can control a base-class complexity term, and why does the training-error product bound not compare two zero-error ensembles?

Hint: Separate a class-capacity notion from an empirical sign certificate.

Answer:

> The order is AdaBoost; its training-error product bound; VC dimension; and the three complementary AdaBoost views. VC dimension supplies the base-class capacity notion. Once two ensembles both have zero training error, the sign count is the same for both, so the product certificate does not record how far their real scores lie from zero.

AdaBoost supplies nonnegative coefficients when each accepted binary weak hypothesis has weighted error below one half. Its registered real score is

$$F_{T(x)} = \sum_{t=1}^{T} \alpha_t h_{t(x)}, \tag{1}$$

and the predicted label is $\operatorname{sign}\left(F_{T(x)}\right)$. Keep those two objects apart: the score is real-valued; the prediction retains only its sign.

For the two-hypothesis score $F(x) = 2h_1(x) + h_2(x)$, complete the blank cells before reading a definition. The coefficient vector is $\alpha = (2, 1)$ and $\|\alpha\|_1 = 3$.

Card	y	$h_1(x)$	$h_2(x)$	$yF(x)$	$\lvert F(x)\rvert$
A	$+1$	$+1$	$+1$		
B	$+1$	$+1$	-1		
C	-1	$+1$	-1		

Candidate card	before $\alpha \to 3\alpha$	after $\alpha \to 3\alpha$	What does it measure?
signed raw score	$yF(x)$		
signed ratio	$y\frac{F(x)}{\lVert\alpha\rVert_1}$		
unsigned ratio	$\lvert F(x)\frac{\rvert}{\lVert\alpha\rVert_1}$		
input-space distance	not computable from this table		

Table 1: Rescaling and competing-quantity cards. Compute first; then label the signed invariant, the unsigned output-space distance, and the quantity this information cannot determine.

The geometric point used by Mohri is not α and not the original input x. It is the weak-hypothesis-output vector $h(x) = \big(h_1(x), \ldots, h_{T(x)}\big)$ in $\mathbb{R}^T$. On your page, write the measured space beside each candidate card and draw a line from $h(x)$ to the hyperplane $\{z \in \mathbb{R}^T : \alpha \cdot z = 0\}$ only on the unsigned card.

IN-CLASS CHECK

Prompt: Compute all three rows, rescale α to $(6, 3)$, and identify what changes. On card C, compare the sign of the invariant signed ratio with the unsigned ratio.

Hint: For the rescaled ensemble, both the numerator and the coefficient norm acquire the same factor.

Answer:

A has $F = 3$, $yF = 3$, and $|F| = 3$; B has $F = 1$, $yF = 1$, and $|F| = 1$; C has $F = 1$, $yF = -1$, and $|F| = 1$. Rescaling gives $F' = 3F$ and $\lVert 3\alpha\rVert_1 = 3\lVert\alpha\rVert_1$, so the raw signed score triples while both ratios retain their magnitudes. Card C has signed ratio $-\frac{1}{3}$ but unsigned ratio $\frac{1}{3}$. The sign disagreement is exactly what prevents the two quantities from being identified on a misclassified example.

Definition: Normalized functional margin

For a labeled example (x, y) and a nonzero coefficient vector α, the course-registered **normalized functional margin** is

$$\rho(x, y) = \frac{yF_{T(x)}}{\lVert\alpha\rVert_1}, \quad \lVert\alpha\rVert_1 = \sum_{t=1}^{T} |\alpha_t|. \tag{2}$$

Its sign records whether the ensemble classifies the example correctly; its magnitude records signed-score confidence after arbitrary positive coefficient scale has been removed. This is the registered form $\rho(x, y) = y\frac{F_{T(x)}}{\lVert\alpha\rVert_1}$.

This normalized signed confidence is not an input-space distance. Mohri's Definition 7.3 instead considers the unsigned quantity

$$d_{\infty(h(x),\{z:\alpha\cdot z=0\})} = \frac{|\alpha \cdot h(x)|}{\|\alpha\|_1} = \frac{|F_{T(x)}|}{\|\alpha\|_1}. \tag{3}$$

The measured point $h(x)$ lies in weak-hypothesis-output space $\mathbb{R}^T$; the coefficient L^1 norm induces the dual L^∞ distance to the α-normal hyperplane. It is not a distance between coefficient vectors. If the example is correctly classified, then $yF_{T(x)} = |F_{T(x)}|$, so this unsigned distance equals $\rho(x, y)$. If the example is misclassified, the distance remains positive while $\rho(x, y) < 0$.

Result: When normalization is a convex combination

If every $\alpha_t \geq 0$ and $\sum_t \alpha_t > 0$, then $\|\alpha\|_1 = \sum_t \alpha_t$ and

$$\overline{F}(x) = \frac{F_{T(x)}}{\|\alpha\|_1} = \sum_{t=1}^{T} \frac{\alpha_t}{\|\alpha\|_1} h_{t(x)} \tag{4}$$

is a convex combination of the base hypotheses. Division by the positive norm preserves $\operatorname{sign}\big(F_{T(x)}\big)$. If signed coefficients are allowed, this convex-hull conclusion requires sign closure of the base class or an explicit reparameterization; the L^1 denominator alone does not supply it.

Worked example: Three cards, three meanings

With $\alpha = (2, 1)$, card A has normalized functional margin 1 and Mohri's unsigned output-space distance 1. Card B has both values $\frac{1}{3}$. Card C has normalized functional margin $-\frac{1}{3}$ but unsigned distance $\frac{1}{3}$. No row determines a Euclidean distance from the original input x to a decision boundary, because neither an input-space metric nor a feature map has been supplied.

IN-CLASS CHECK

Prompt: A signed-coordinate variant uses $\alpha = (2, -1)$ with the original dictionary $\{h_1, h_2\}$. Which statements remain true, and which convex-hull statement needs repair?

Hint: Separate positive scale invariance from membership in $\operatorname{conv}(\mathcal{H})$.

Answer:

> Dividing by the positive number $\|\alpha\|_1 = 3$ still preserves the sign of the real score, and $y\frac{F}{\|\alpha\|_1}$ remains invariant under a common positive rescaling. But the normalized coefficients $(\frac{2}{3}, -\frac{1}{3})$ are not convex weights over the original dictionary. To invoke a convex-hull theorem, the class must be closed under sign so that $-h_2$ becomes a permitted coordinate, or the representation must be changed explicitly.

The scale question is now settled for one example. The next loss of information occurs when many normalized margins are collapsed back to a single zero-error count.

Let the threshold move across the training sample

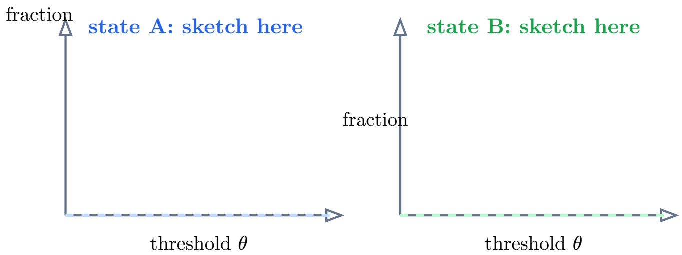
Consider two later AdaBoost states evaluated on the same five training examples. Both states classify every example correctly. Their normalized margins are

$$A = (0.05, 0.10, 0.25, 0.65, 0.90), \quad B = (0.20, 0.40, 0.55, 0.75, 0.95). \tag{5}$$

The zero-error count sees no difference. A moving threshold does.

Before naming the resulting curve, fill every blank in the ledger. Each entry is the fraction of the five margins at or below the threshold in the first column.

Threshold θ	**State A**	**State B**	**Must the fraction fall as θ grows?**
0			
0.10			
0.30			
0.60			
0.80			
1.00			



validation trace decreases	validation trace stays flat	validation trace increases
compatible / impossible?	compatible / impossible?	compatible / impossible?

Table 2: Threshold ledger, two blank step-curve axes, and three candidate population traces. Training margins determine the first two objects, not the third.

IN-CLASS CHECK

Prompt: Complete the table and curves. Which state has less mass at or below $\theta = 0.30$? Mark every candidate validation trace that remains compatible with the displayed training evidence.

Hint: At a larger threshold, the set of counted examples can only grow. Population behavior has not been observed.

Answer:

For A the six fractions are $0, \frac{2}{5}, \frac{3}{5}, \frac{3}{5}, \frac{4}{5}, 1$; for B they are $0, 0, \frac{1}{5}, \frac{3}{5}, \frac{4}{5}, 1$. Thus B has less mass at or below 0.30. Each curve is a nondecreasing right-continuous step curve under the at-or-below convention. Decreasing, flat, and increasing validation traces all remain compatible because the table contains training evidence only.

Definition: Empirical margin distribution

For an i.i.d. training sample $S = ((x_i, y_i))_{i=1}^m$ and threshold θ, the **empirical cumulative low-margin fraction** is

$$\hat{R}_{S,\theta}(\overline{F}) = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}[\rho(x_i, y_i) \leq \theta]. \tag{6}$$

Varying θ gives the empirical margin distribution. It is cumulative: if $\theta_1 \leq \theta_2$, then $\hat{R}_{S,\theta_1} \leq \hat{R}_{S,\theta_2}$.

At threshold zero, and away from a zero-score tie convention, this fraction recovers the empirical sign error. Positive thresholds retain more information: they count examples that are correct but still close to zero on the normalized score scale. The word **empirical** matters. A right-shifted training-margin curve can accompany lower test error, as observed in some boosting experiments, but the curve alone does not establish a population trajectory.

Worked example: Read one threshold, not a whole story

At $\theta = 0.30$, state A has empirical low-margin mass $\frac{3}{5}$, whereas state B has mass $\frac{1}{5}$. This is a precise training comparison. It does not say that B has test error $\frac{1}{5}$, that B's test error is below A's, or that another boosting round will continue the same shift.

IN-CLASS CHECK

Prompt: A learner says, "B's curve lies below A's at $\theta = 0.30$, so B must generalize better." Repair the statement without discarding the valid observation.

Hint: Name the sample on which the curve is computed and the missing bridge to population risk.

Answer:

> The valid statement is that, on this fixed training sample, B has a smaller fraction of normalized margins at or below 0.30. A population comparison needs a theorem with a declared function class, sample model, confidence level, threshold, and complexity term, or separate validation evidence.

We now have the empirical term that such a theorem could use. The remaining work is to expose every condition before trusting the population side.

Build the population certificate around the empirical curve

The cards below are deliberately detached. Work from left to right before the formal statement appears.

Candidate ensemble	Admit to normalized convex class?	Reason
$0.7h_1 + 0.3h_2$		
$2h_1 + h_2$ as written		
$0.7h_1 - 0.3h_2$ without sign closure		

Wrapper order	Card 1	Card 2	Card 3
write 1, 2, 3	fix $\rho_0 > 0$	draw i.i.d. sample S	with probability $\geq 1 - \delta$, for every normalized ensemble

population zero-one error	empirical ρ_0-low-margin mass	base-class complexity divided by ρ_0	sample/confidence term
left or right?	left or right?	left or right?	left or right?

Table 3: A class gate, a probability-wrapper ordering task, and detached bound terms. Fix the threshold before drawing the confidence wrapper.

IN-CLASS CHECK

Prompt: Reject the invalid class members, order the three wrapper cards, and assemble an inequality. Then explain why writing "for all ρ_0" inside the same wrapper needs an additional correction.

Hint: Convex weights are nonnegative and sum to at most one; the displayed core claim fixes one positive threshold before its probability statement.

Answer:

The first ensemble is admitted. The second must first be normalized to $(\frac{2}{3})h_1 + (\frac{1}{3})h_2$. The third is not in the original convex hull without sign closure or reparameterization. The order is: fix $\rho_0 > 0$; draw the i.i.d. sample; then, with probability at least $1 - \delta$, assert the bound for every function in the normalized class. Population error belongs on the left; empirical low-margin mass and the separate complexity and confidence terms belong on the right. Making the statement simultaneous over every threshold in $(0, 1]$ requires Mohri's additional log-log correction, which is absent from this fixed-threshold assembly.

Definition: Normalized convex ensemble class at a fixed margin threshold

For a binary base class $\mathcal{H}$,

$$\operatorname{conv}(\mathcal{H}) = \{\sum_{j=1}^{p} \mu_j h_j : p \geq 1, \mu_j \geq 0, h_j \in \mathcal{H}, \sum_{j=1}^{p} \mu_j \leq 1\}. \tag{7}$$

The core theorem evaluates every function in this normalized convex class at one fixed positive threshold ρ_0. An arbitrary real linear combination is not automatically a member. An AdaBoost score with nonnegative coefficients enters after division by its positive L^1 coefficient norm.

Why can the complexity term depend on the base class rather than the number of selected rounds? Mohri's **Lemma 7.4** states that taking the convex hull does not increase empirical Rademacher complexity. Its decisive algebraic step is that a linear expression over convex weights is maximized at an extreme point: all available weight can be placed on the best base hypothesis. This is a property of the normalized class. It is not a theorem that additional boosting rounds improve an observed curve.

Theorem: Fixed-threshold boosting margin bound

Let $\mathcal{H}$ be a class of functions from $\mathcal{X}$ to $\{-1, +1\}$ with $d = \mathrm{VCdim}(\mathcal{H})$ and $1 \leq d \leq m$. Fix $\rho_0 > 0$ and $\delta \in (0, 1)$. For an i.i.d. sample S of size m, with probability at least $1 - \delta$, the following holds simultaneously for every $\overline{F} \in \operatorname{conv}(\mathcal{H})$:

$$R(\overline{F}) \leq \hat{R}_{S,\rho_0}(\overline{F}) + \frac{2}{\rho_0}\sqrt{\frac{2d\log(e\frac{m}{d})}{m}} + \sqrt{\frac{\log(\frac{1}{\delta})}{2m}}. \tag{8}$$

Here $R(\overline{F}) = \Pr\left[y\overline{F}(x) \leq 0\right]$ is population zero-one error, while $\hat{R}_{S,\rho_0}$ is the empirical fraction whose registered normalized margin is at most ρ_0. This is the VC-dimension form of Mohri Corollary 7.6; related Rademacher forms appear in Corollary 7.5.

Read the quantifiers in the order in which they were assembled. The threshold is fixed before the sample-probability statement. The probability is over the i.i.d. sample. Once the high-probability event holds, the inequality is uniform over functions in the normalized class. It is not, in this displayed form, uniform over threshold choices.

Each term has one job:

- $\hat{R}_{S,\rho_0}$ is observed low-margin training mass;
- $d = \mathrm{VCdim}(\mathcal{H})$ is base-class capacity;
- $\frac{1}{\rho_0}$ charges for asking that margins clear a larger threshold;
- m supplies sample-size decay; and
- δ controls the failure probability.

No term is an observed test error, and the inequality is not an equality.

Worked example: Two separately instantiated thresholds

Hold the class, sample size, VC dimension, and confidence level fixed, and abbreviate the displayed complexity numerator by $C = 0.04$ and the final confidence term by $q = 0.03$. Suppose the empirical curve gives $\hat{R}_{S,0.20} = 0.10$ and $\hat{R}_{S,0.40} = 0.25$. The two fixed-threshold certificates are

$$0.10 + \frac{0.04}{0.20} + 0.03 = 0.33, \quad 0.25 + \frac{0.04}{0.40} + 0.03 = 0.38. \tag{9}$$

A larger threshold improves the inverse-threshold term but can worsen the empirical term. These are two separate theorem instantiations, not a claim that the displayed confidence event was already uniform over every threshold.

IN-CLASS CHECK

Prompt: In the worked comparison, increase the sample size by a factor of four while holding the displayed empirical fractions, d, δ, and both thresholds fixed. Which terms shrink, and by what qualitative factor?

Hint: Both nonempirical square-root terms contain m in the denominator; keep the logarithmic dependence visible.

Answer:

The VC term and the confidence term both shrink on an approximate $\frac{1}{\sqrt{m}}$ scale, so the leading square-root effect is a factor of about one half. The VC logarithm also changes from $\log(\frac{em}{d})$ to $\log(\frac{4em}{d})$, so its reduction is not exactly one half. The empirical fractions need not change under the stipulated comparison because they were explicitly held fixed.

IN-CLASS CHECK

Prompt: Audit four claims: (i) the theorem has no explicit T; (ii) more rounds must improve the empirical margin curve; (iii) more rounds must reduce test error; (iv) the theorem upper bound equals observed population error. Which claim is licensed?

Hint: Distinguish a formula's parameter dependence from a trajectory theorem.

Answer:

Only (i) is licensed: after positive-coefficient normalization, the displayed bound has no explicit round-count term. Claims (ii) and (iii) require evidence about the actual training and population trajectories. Claim (iv) is false because the theorem gives an upper bound, not an equality or an observation of population error.

IN-CLASS CHECK

Prompt: A learner chooses the most favorable threshold after seeing the sample and then quotes the displayed fixed-threshold confidence statement unchanged. What is missing?

Hint: The missing issue is threshold uniformity, not function uniformity.

Answer:

The displayed event is uniform over normalized ensemble functions only after one threshold has been fixed. A statement simultaneous over all $\rho_0 \in (0, 1]$ requires the additional log-log correction described by Mohri. Without that corrected theorem, post-sample threshold selection cannot inherit the same confidence statement automatically.

The theorem has now connected the empirical curve to a population quantity, but only conditionally. Its normalized class, fixed threshold, i.i.d. sample, confidence level, and base-class capacity are part of the conclusion, not fine print.

Standard language to carry forward

Term or notation	Meaning here	Representative form
normalized functional margin	Signed confidence after removing common positive coefficient scale.	$\rho(x, y) = y\frac{F_{T(x)}}{\|\alpha\|_1}$
L^1-normalized boosting ensemble	A positive-coefficient AdaBoost score divided by its coefficient sum; a convex combination with unchanged sign.	$\overline{F} = \frac{F_T}{\|\alpha\|_1}$
empirical margin distribution	Cumulative training fraction at or below a threshold.	$\hat{R}_{S,\theta} = m^{-1} \sum_i \mathbf{1}[\rho_i \leq \theta]$
normalized convex ensemble class	Convex combinations of base hypotheses to which the fixed-threshold theorem applies.	$\mathrm{conv}(\mathcal{H})$
boosting margin generalization bound	Population zero-one error is bounded by empirical low-margin mass plus distinct complexity and confidence terms.	$R \leq \hat{R}_{S,\rho_0} + \text{complexity} + \text{confidence}$

Reconstruct the exact Tutorial 7B entry

Tutorial 7B may use exactly ten items from this boundary, in order. The first four are the Chapter 6 entry just reconstructed; the next five were established in this Tutorial; the last is the registered symbolic form.

No.	Available item	What Tutorial 7B may ask you to do
1	AdaBoost	Reconstruct the normalized update and real score.
2	AdaBoost training-error product bound	Separate the empirical certificate from population evidence.
3	VC dimension	Identify base-class capacity in the margin theorem.
4	three complementary AdaBoost views	Match a claim to the evidence its view supports.
5	normalized functional margin	Compute signed confidence at a common coefficient scale.
6	L^1 ensemble confidence normalization	State the positive-coefficient convex-combination condition and signed-coordinate caveat.
7	empirical margin distribution	Compute and interpret a cumulative low-margin fraction.

No.	Available item	What Tutorial 7B may ask you to do
8	normalized convex ensemble class	Check membership before invoking the theorem.
9	boosting margin generalization bound	State its fixed-threshold quantifiers and distinguish every term.
10	$\rho(x, y) = y\frac{F_{T(x)}}{\|\alpha\|_1}$	Use the registered notation without replacing it by an input-space distance.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct all ten items in order. Which item carries a population certificate, and which item records training evidence but cannot by itself supply one?

Hint: Keep item 7 and item 9 separate.

Answer:

The order is AdaBoost; the AdaBoost training-error product bound; VC dimension; the three complementary AdaBoost views; normalized functional margin; L^1 ensemble confidence normalization; empirical margin distribution; normalized convex ensemble class; boosting margin generalization bound; and $\rho(x, y) = y\frac{F_{T(x)}}{\|\alpha\|_1}$. Item 9 supplies the conditional population certificate. Item 7 is empirical training evidence and needs item 8 plus the sample, confidence, threshold, and complexity conditions of item 9.

The remaining question is deliberately unanswered here: which empirical objective and which coordinate rule reproduce AdaBoost's update exactly? Tutorial 7B will define those objects before using them.

Source note

The historical Week 7 learning sheet fixes the scope of the three source units used here: the post-zero margin question, the reconstructed functional-margin definition, and the population margin bound. Mathematical authority comes from Mohri, Rostamizadeh, and Talwalkar, Sections 7.3.2-7.3.3, especially Definition 7.3, Lemma 7.4, Corollaries 7.5-7.6, and equations (7.10)-(7.17), together with Schapire, **Explaining AdaBoost**, Section 3. The geometric point is $h(x)$ in weak-hypothesis-output space, not α or the original input. The displayed population theorem fixes one positive threshold before the probability claim; the simultaneous all-threshold version needs the additional log-log correction. No claim here says that more rounds monotonically improve margins or test error.

DSAA3073: Theories in Data Science

Tutorial 7B: AdaBoost as Optimization - and Where That View Stops

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-31

Explorative projection

LESSON BACKBONE

The margin theorem controls population error under its own conditions. Which empirical objective generates AdaBoost's exact update, when is that update a coordinate step, and why does neither fact certify observed test behavior?

Tutorial 7A ended with a population certificate built from a normalized margin distribution. We now return to the training calculation. First we prove that exponential loss dominates the zero-one indicator pointwise. Then we ask a more exact question: among many possible ways to reduce that loss, which coordinate rule and which line search reproduce one AdaBoost round?

The answer is conditional. A fixed binary dictionary, minimum current weighted error, and exact line search are all part of the equivalence. Once those conditions are visible, persistent hard examples reveal why an exact empirical mechanism can still require a separate margin theorem and held-out evidence.

The three core blocks take **45, 60, and 45 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-45	Prove the exponential surrogate bound and delimit what it says	45 min core
45-60	Scheduled break	15 min
60-120	Derive the coordinate direction, line minimum, and exact-equivalence conditions	60 min core
120-165	Diagnose persistent hard examples and match claims to evidence	45 min core
165-225	Questions, proof reconstruction, alternative calculations	60 min buffer

Replace a difficult target without replacing the question

Tutorial 7A made exactly ten items available at this boundary, in order. The first four came through Chapter 6; the next five were established in Tutorial 7A; the last is the registered symbolic form.

No.	Available item	Use in this Tutorial
1	AdaBoost	Reconstruct the normalized example-weight update and real score.
2	AdaBoost training-error product bound	Keep an empirical certificate distinct from population evidence.
3	VC dimension	Recognize the base-class capacity term in the margin theorem.
4	three complementary AdaBoost views	Ask which evidence can support a proposed claim.
5	normalized functional margin	Track signed confidence after removing common coefficient scale.
6	L^1 ensemble confidence normalization	Recall the positive-coefficient condition and signed-coordinate caveat.
7	empirical margin distribution	Treat low-margin mass as training evidence.
8	normalized convex ensemble class	Check theorem membership before using the certificate.
9	boosting margin generalization bound	Keep its fixed-threshold conditions and probability scope intact.
10	$\rho(x, y) = y\frac{F_{T(x)}}{\|\alpha\|_1}$	Use the registered signed form without turning it into input-space distance.

IN-CLASS CHECK

Prompt: Reconstruct all ten items in order. Which two items most directly prevent an empirical optimization claim from becoming an automatic population conclusion?

Hint: One records a theorem with explicit population conditions; the other tells you that different explanations require different evidence.

Answer:

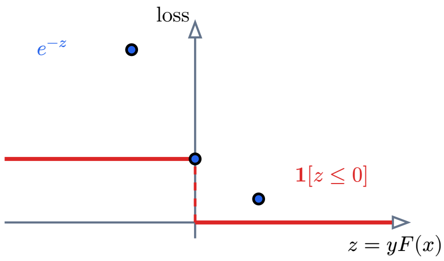
The order is AdaBoost; the training-error product bound; VC dimension; the three complementary views; normalized functional margin; L^1 confidence normalization; empirical margin distribution; normalized convex ensemble class; the boosting margin bound; and $\rho(x, y) = y\frac{F_{T(x)}}{\|\alpha\|_1}$. Item 9 supplies a conditional population theorem, while item 4 prevents the empirical optimization view from silently inheriting that theorem's conclusion.

The ten rows above remain the exact Tutorial 7A handoff. Separately, we recall Chapter 1's registered meaning of ***L*-smoothness**: the derivative must be globally L-Lipschitz on the declared domain. The property sort below uses that precise definition; merely possessing derivatives of every order is a different statement.

For a labeled example (x, y) with $y \in \{-1, +1\}$ and a real ensemble score $F(x)$, the classification target is the zero-one loss

$$\ell_{0-1}(y, F(x)) = \mathbf{1}[yF(x) \le 0]. \tag{10}$$

The registered exponential loss from Chapter 6 is $\ell_{\exp}(y, F(x)) = e^{-yF(x)}$. Before attaching a general label to either function, use the signed score $z = yF(x)$ to complete the visual below.



Card	Decide now	Evidence needed	Do not merge with
pointwise domination	yes / no	two sign cases	convexity
convex objective	yes / no	composition with an affine score	differentiability
differentiable / C^∞ objective	yes / no	derivatives of e^{-z}	global L-smoothness
globally L-smooth objective on $\mathbb{R}$	yes / no	a uniform bound on $\varphi''(z)$	bounded-interval smoothness
population guarantee	yes / no	a separate generalization theorem	an empirical average
robustness to label noise	yes / no	data, weak class, and validation evidence	pointwise domination

Table 4: Signed-score strip and property cards. Fill the relation and sort the claims before the formal surrogate role is named.

IN-CLASS CHECK

Prompt: Fill the missing relation for both $z \le 0$ and $z > 0$, then sort the six cards. Which fact is sufficient for the relation?

Hint: When $z \le 0$, compare two quantities that are both at least one. When $z > 0$, inspect the indicator first.

Answer:

If $z \le 0$, then $\mathbf{1}[z \le 0] = 1$ and $e^{-z} \ge 1$. If $z > 0$, then the indicator is zero and $e^{-z} > 0$. Hence $\mathbf{1}[z \le 0] \le e^{-z}$ in both cases. This two-case argument proves pointwise domination. Exponential loss is also convex and infinitely differentiable, but it is not globally L-smooth on $\mathbb{R}$ because $\varphi''(z) = e^{-z}$ is unbounded as $z \to -\infty$. On $z \in [-B, B]$, it is L-smooth with $L = e^B$. Population performance, calibration, and robustness each require their own statement and evidence.

Definition: Convex classification surrogate

A **convex classification surrogate** replaces the difficult zero-one training objective by a convex real-valued objective. Its optimization properties and its statistical relationship to the classification target must be stated separately. Convexity, differentiability, global or bounded-domain L-smoothness, pointwise domination, calibration, and robustness are not interchangeable names for the same property.

The exponential loss is the surrogate used by AdaBoost. The visual already contains the proof of its basic pointwise relationship.

Result: Exponential loss upper-bounds binary zero-one loss

For every $y \in \{-1, +1\}$ and every real score $F(x)$,

$$\mathbf{1}[yF(x) \leq 0] \leq e^{-yF(x)}. \tag{11}$$

Therefore, on a fixed sample $S = ((x_1, y_1), \ldots, (x_m, y_m))$,

$$\frac{1}{m}\sum_{i=1}^{m} \mathbf{1}[y_i F(x_i) \leq 0] \leq \frac{1}{m}\sum_{i=1}^{m} e^{-y_i F(x_i)}. \tag{12}$$

The consequence is an empirical average. No population or calibration claim follows from this averaging step alone.

IN-CLASS CHECK

Prompt: A model has empirical exponential loss 0.08. What can you conclude about its empirical zero-one error, and what can you not conclude about its test error?

Hint: Average the pointwise inequality; do not change an empirical average into an expectation over new data.

Answer:

> Its empirical zero-one error is at most 0.08. This says nothing automatic about the observed test error or population risk. A valid population claim still needs a generalization theorem with its class, sample, confidence, and other conditions, or held-out evidence for the fitted model.

The empirical exponential objective is now legitimate, but an objective does not determine a unique optimization path. That distinction is the reason to derive AdaBoost's coordinate and step instead of merely calling the loss convex.

Make one coordinate step reproduce one AdaBoost round

Fix a finite dictionary of binary base hypotheses $\mathcal{H} = \{h_1, ..., h_N\}$ with $h_{k(x)} \in \{-1, +1\}$. A coefficient vector $\beta \in \mathbb{R}^N$ defines $F_{\beta(x)} = \sum_{j=1}^{N} \beta_j h_{j(x)}$ and the empirical exponential objective

$$\mathcal{L}(\beta) = \frac{1}{m} \sum_{i=1}^{m} \exp\left(-y_i F_{\beta(x_i)}\right). \tag{13}$$

At the start of round t, let $F_{t-1} = F_{\beta^{t-1}}$ and normalize the current exponential masses:

$$D_{t(i)} = \frac{e^{-y_i F_{t-1}(x_i)}}{\sum_{r=1}^{m} e^{-y_r F_{t-1}(x_r)}}. \tag{14}$$

For the lab below, the displayed distribution is $(0.40, 0.30, 0.20, 0.10)$. A plus sign means $h_{k(x_i)} = y_i$ and a minus sign means $h_{k(x_i)} \neq y_i$.

Index	$D_{t(i)}$	h_1	h_2	h_3
1	0.40	+	+	−
2	0.30	+	−	+
3	0.20	−	+	+
4	0.10	−	−	+
weighted error				
derivative / common factor				

Case	Coordinate rule	Line rule	Classification
A	minimum D_t-weighted error	exact minimum	
B	approximate weak learner	exact minimum	
C	minimum weighted error	inexact step	
D	cyclic coordinate	exact minimum	
E	another optimizer of $\mathcal{L}$	its own update	

IN-CLASS CHECK

Prompt: Compute the three weighted errors. If the directional derivative is a positive common factor times $2\varepsilon_{t,k} - 1$, which coordinate gives the greatest immediate decrease?

Hint: Add the weights in the minus rows for each column. The most negative derivative is the descent choice.

Answer:

> The errors are $\varepsilon_{t,1} = 0.20 + 0.10 = 0.30$, $\varepsilon_{t,2} = 0.30 + 0.10 = 0.40$, and $\varepsilon_{t,3} = 0.40$. The derivative factors are therefore $-0.40, -0.20$, and -0.20. Coordinate 1 has the most negative derivative and gives the greatest immediate decrease. Weighted error is doing the selection work; the unweighted number of mistakes would not distinguish these columns correctly.

Definition: Coordinate descent - the variant used here

Coordinate descent changes one coefficient while holding all other coefficients fixed. In this Tutorial, the rule chooses a maximum-descent coordinate from the fixed finite binary dictionary and then minimizes the objective exactly along that one coordinate. Cyclic, random, approximate, and block-coordinate rules are different variants.

To see why weighted error selects the direction, move from the current score along coordinate k: $F_\eta = F_{t-1} + \eta h_k$. Write $C_t = \mathcal{L}(\beta^{t-1}) > 0$. Differentiating at $\eta = 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}\eta}\mathcal{L}(F_{t-1} + \eta h_k) \mid_{\eta=0} = -\frac{1}{m}\sum_i y_i h_{k(x_i)} e^{-y_i F_{t-1}(x_i)} = C_{t(2\varepsilon_{t,k}-1)}, \tag{15}$$

where $\varepsilon_{t,k} = \sum_i D_{t(i)} \mathbf{1}\left[h_{k(x_i)} \neq y_i\right]$. The factor C_t is positive and independent of k. Thus, among coordinates with error below one half, minimizing the **current weighted error** is exactly the same as choosing the most negative directional derivative.

IN-CLASS CHECK

Prompt: Derive the final expression by splitting the normalized sum into correct and incorrect mass. Why may C_t be ignored when comparing coordinates but not declared equal to one?

Hint: On correct rows $y_i h_{k(x_i)} = +1$; on incorrect rows it is -1.

Answer:

> The normalized signed sum is $\sum_i D_{t(i)} y_i h_{k(x_i)} = \left(1 - \varepsilon_{t,k}\right) - \varepsilon_{t,k} = 1 - 2\varepsilon_{t,k}$. The derivative is its negative times C_t, hence $C_{t(2\varepsilon_{t,k}-1)}$. The factor is common to every coordinate, so it does not change their ordering. It is the current empirical exponential loss, not generally one, and must remain in an equality for the derivative.

After coordinate k is selected, split the one-dimensional objective into the D_t-mass classified correctly and incorrectly:

$$\mathcal{L}(F_{t-1} + \eta h_k) = C_{t((1-\varepsilon_t)e^{-\eta} + \varepsilon_t e^{\eta})}. \tag{16}$$

For $0 < \varepsilon_t < \frac{1}{2}$, its second derivative is positive, and the first-order condition is

$$-(1-\varepsilon_t)e^{-\eta} + \varepsilon_t e^{\eta} = 0. \tag{17}$$

Therefore the unique finite minimizer is

$$\eta = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right) = \alpha_t > 0. \tag{18}$$

For the lab coordinate, $\varepsilon_t = 0.30$, so $\alpha_t = \frac{1}{2}\log\left(\frac{7}{3}\right) \approx 0.4236$. The line-objective factor falls from one at $\eta = 0$ to $2\sqrt{0.30(0.70)} \approx 0.9165$ at the minimum.

IN-CLASS CHECK

Prompt: Starting from the correct/error mass split, recover α_t and the minimum factor. Why is convexity used here?

Hint: Set the derivative to zero, then substitute the resulting square-root ratios.

Answer:

The stationary equation gives $e^{2\eta} = \frac{1-\varepsilon_t}{\varepsilon_t}$, hence $\eta = \alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$. Substitution yields $(1-\varepsilon_t)\sqrt{\frac{\varepsilon_t}{1-\varepsilon_t}} + \varepsilon_t\sqrt{\frac{1-\varepsilon_t}{\varepsilon_t}} = 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$. Convexity makes this stationary point the global line minimum. It does not prove that every coordinate rule follows the same path or ends at the same classifier.

The three boundary regimes cannot be hidden inside the finite positive formula.

Weighted error	Line objective	Best step	Meaning
$\varepsilon_t = 0$	$e^{-\eta}$	$\eta \to +\infty$	No finite log-odds minimizer.
$\varepsilon_t = \frac{1}{2}$	$\frac{1}{2}(e^{-\eta} + e^{\eta})$	$\eta = 0$	Zero step; no positive progress.
$\frac{1}{2} < \varepsilon_t < 1$	convex correct/error split	$\eta < 0$	A negative direction; use only with sign closure or explicit reparameterization.

IN-CLASS CHECK

Prompt: Classify the five lab cases A-E, then add the three error-boundary rows. Which cases are exact AdaBoost-coordinate steps?

Hint: Exact means both the coordinate and line rules match, with a finite positive coefficient under $0 < \varepsilon_t < \frac{1}{2}$.

Answer:

Case A is exact when its selected error lies strictly between zero and one half. B is an approximate greedy coordinate step, C has the right coordinate but not the exact AdaBoost coefficient, D is coordinate descent but generally not AdaBoost, and E shares an objective but not necessarily a path, classifier, or test behavior. At error zero the optimum is reached only as the coefficient tends to infinity; at error one half the step is zero. An error above one half gives a negative step and may be converted to a positive step on $-h_k$ only when the dictionary is sign-closed, or when an explicit equivalent reparameterization has been declared.

Result: Conditional AdaBoost-coordinate-descent equivalence

Consider binary base hypotheses from a fixed finite coordinate dictionary (or a separately defined oracle analogue), the empirical exponential objective, and the current normalized exponential distribution D_t. If the weak learner selects a minimum-D_t-weighted-error coordinate, if $0 < \varepsilon_t < \frac{1}{2}$, and if the line search is exact, then maximum-descent coordinate selection returns h_t and the line minimum returns the finite positive AdaBoost coefficient $\alpha_t = \frac{1}{2}\log\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$. Hence that one greedy coordinate step coincides with one AdaBoost round.

The zero-error, half-error, and negative-direction cases require the boundary treatment above. Approximate selection or line search gives only an approximate greedy step. Generic coordinate descent or another optimizer of exponential loss need not reproduce AdaBoost's path, classifier, margins, or generalization behavior.

We have proved a precise empirical mechanism. The remaining issue is not a gap in the derivative. It is a different question: what happens when the weights keep concentrating on examples that the declared weak class cannot resolve?

Read a persistent hard example without guessing its cause

The two traces below are illustrative normalized training-weight histories. Example M is later found, by an external label audit, to be mislabeled. Example C is clean but difficult for the available weak-hypothesis class. Until that audit is opened, only the numerical traces are visible.

Trace	D_1	D_2	D_3	D_4	**What the trace alone proves**
example M	0.05	0.14	0.31	0.47	
example C	0.05	0.13	0.30	0.46	
all other examples	0.90	0.73	0.39	0.07	remaining normalized mass

Proposed claim	**Relevant view**	**Evidence that can support it**	**What that evidence cannot replace**
round mechanics	example reweighting	D_t update and a normalized trace	a population theorem
objective decrease	optimization	exponential objective, direction, and line calculation	AdaBoost path under another update rule
population certificate	margin	fixed-threshold theorem and all its conditions	observed test performance
fitted model's test behavior	external validation	untouched held-out or deployment evidence	a proof of universal behavior

IN-CLASS CHECK

Prompt: Before opening the label audit, what do the two traces establish? Why can neither trace identify label noise by itself?

Hint: Both examples can stay wrong for the current weak class, but for different reasons.

Answer:

> Each trace establishes persistent concentration of normalized training weight on one hard example. Because $D_t(i)$ is proportional to $e^{-y_i F_{t-1}(x_i)}$, repeated negative or small signed scores can increase its relative mass. The same pattern can be produced by a wrong label, a rare but correctly labeled region, an insufficient weak class, or a still-early optimization state. Label noise is therefore not identifiable from a large weight alone.

Definition: AdaBoost sensitivity to persistent hard examples

A mislabeled or persistently hard example can retain large exponential weight and influence later weak-hypothesis choices. This is a conditional risk, especially when the available weak class is insufficiently expressive. It is not a universal noise threshold, a proof that every high-weight label is wrong, or a theorem that test error must worsen monotonically with rounds.

The optimization derivation explains the concentration mechanism: a negative signed score produces a large exponential factor, and the weak learner sees the result through D_t. It does not diagnose the data-generating cause. Schapire's noise discussion makes both sides important: under suitable richness and stopping conditions AdaBoost can be consistent, while insufficient weak-class expressiveness can make loss-based methods very poor even with mild constructed noise. The responsible conclusion is conditional, not a fixed percentage at which one method becomes safe or unsafe.

IN-CLASS CHECK

Prompt: Repair these four claims: (i) a large $D_{t(i)}$ proves mislabeling; (ii) low empirical exponential loss certifies low test error; (iii) larger empirical margins guarantee the next fitted model has lower observed test error; (iv) any optimizer of exponential loss is AdaBoost.

Hint: For each sentence, name the missing condition or evidence rather than replacing it with another absolute claim.

Answer:

(i) A large weight shows persistent current difficulty; label status needs external audit or domain evidence. (ii) Low empirical exponential loss upper-bounds empirical zero-one error, while population or test behavior needs a valid generalization theorem or held-out evidence. (iii) A better empirical margin distribution may tighten a fixed-threshold bound when its class, sample, threshold, and confidence conditions hold; it does not force a monotone realized test trace. (iv) AdaBoost is the particular greedy minimum-weighted-error coordinate rule with the declared exact line search and boundary conditions. Another optimizer can reach different coefficients, margins, classifier, and test behavior.

Definition: Selecting evidence for an AdaBoost claim

Use example weights to explain update mechanics, exponential loss to justify the declared empirical optimization step, the boosting margin theorem for its conditional population certificate, and held-out evidence for the observed test behavior of a fitted model. None of these four evidence types automatically supplies the others' conclusion.

IN-CLASS CHECK

Prompt: A report says: "Both algorithms reached empirical exponential loss 0.03, so their classifiers and test errors are effectively the same." Which evidence column supports the premise, and what must replace the conclusion?

Hint: The premise concerns an objective value; the conclusion contains both a path-dependent classifier claim and an observed-performance claim.

Answer:

> The optimization column can verify the common empirical objective value. It does not identify the coefficient vectors, signs, normalized margins, or update paths. Compare the actual fitted classifiers to support a classifier claim, check the margin theorem separately if a conditional population certificate is desired, and use untouched held-out data to compare observed test performance.

Standard language to carry forward

Term	Meaning here	Representative form or condition
convex classification surrogate	A tractable objective whose optimization and statistical links to zero-one loss are stated separately.	convexity does not imply every other property
exponential upper bound	A pointwise inequality and its empirical-average consequence.	$\mathbf{1}[yF \le 0] \le e^{-yF}$
coordinate descent	A one-coefficient update; the variant here uses maximum descent and exact line search.	$\beta \to \beta + \eta e_k$
AdaBoost as greedy coordinate descent	An exact conditional equivalence, not a name for every exponential-loss optimizer.	fixed binary dictionary, minimum weighted error, exact line search, $0 < \varepsilon_t < \frac{1}{2}$
sensitivity to persistent hard examples	Exponential weights can concentrate on unresolved examples under stated data and weak-class conditions.	large weight does not identify a wrong label
evidence selection	Mechanics, optimization, a population theorem, and observed validation answer different questions.	state the claim before choosing the evidence

Close Chapter 7 and reconstruct the Chapter 8 entry

Chapter 7 closes with three results that remain part of the course record:

1. the boosting margin generalization bound, with its fixed-threshold conditions;
2. the conditional AdaBoost-coordinate-descent equivalence;
3. the normalized functional margin $\rho(x, y) = y\frac{F_{T(x)}}{\|\alpha\|_1}$.

Those chapter exits are not the same as the next chapter's prerequisite list. Chapter 8 receives exactly three earlier tools, in order:

No.	Available item	What Chapter 8 may reuse
1	Hoeffding concentration	Control a bounded independent average around its expectation.
2	normalized example-distribution pattern	Maintain and interpret nonnegative weights that sum to one.
3	AdaBoost product-bound proof pattern	Track a multiplicative potential through local normalization factors.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct the three Chapter 7 exits and then the exact three-item Chapter 8 entry. Why are the two lists intentionally different?

Hint: The exits record what this chapter established; the entry records only what the next chapter actually requires.

Answer:

> The Chapter 7 exits are the boosting margin generalization bound, the conditional AdaBoost-coordinate-descent equivalence, and the normalized functional margin. Chapter 8 receives, in order, Hoeffding concentration; the normalized example-distribution pattern; and the AdaBoost product-bound proof pattern. The first list preserves this chapter's main results. The second is a minimal prerequisite interface: Chapter 8 need not consume the margin theorem or coordinate equivalence merely because they were taught most recently.

Optional deep dive: compare three classification surrogates

0 class minutes; post-class or self-study. Unassessed, non-prerequisite, and not exported to Chapter 8.

Question. Along the signed-score axis $z = yF(x)$, how do hinge, normalized logistic, and exponential losses differ without implying that their algorithms or robustness properties are interchangeable?

Answer. Use the following comparison only for the functions shown:

Loss	Formula	Convex	Differentiability	Global L-smoothness	Negative-margin emphasis
hinge	$\max(0, 1-z)$	yes	not at $z = 1$	no	linear once active
normalized logistic	$\frac{\log(1+e^{-z})}{\log(2)}$	yes	C^∞	yes: $L = \frac{1}{4\log 2}$	approximately linear far left
exponential	e^{-z}	yes	C^∞	no on $\mathbb{R}$; $L = e^B$ on $[-B, B]$	grows exponentially far left

All three displayed scalings upper-bound $\mathbf{1}[z \leq 0]$. The table establishes shape, convexity, differentiability, domain-qualified L-smoothness, and tail emphasis only. For normalized logistic loss, $\varphi''(z) \leq \frac{1}{4\log 2}$ on $\mathbb{R}$; for exponential loss, $\varphi''(z) = e^{-z}$ has no finite global bound. These facts do not prove that one loss is universally more robust, calibrated under every convention, or paired with the same update. In particular, replacing exponential loss changes the line objective, so the exact AdaBoost coefficient derived above is no longer licensed without a new derivation.

Source note

The historical Week 7 learning sheet fixes the scope of the exponential-loss, coordinate-descent, and synthesis units; its blanket efficiency, convergence, noise-threshold, and view-equivalence claims are not used. Mathematical authority comes from Mohri, Rostamizadeh, and Talwalkar, Section 4.7 and Section 7.2.2, especially equations (7.6)-(7.7) and the coordinate derivative and line-search derivation on printed pages 151-153. Schapire, **Explaining AdaBoost**, Sections 4-7 supplies the warning that equal exponential loss does not force equal generalization behavior and the conditional boundary created by noise and insufficient weak-class expressiveness. The optional loss comparison above uses Chapter 1′s registered global L-smoothness definition; it has zero protected minutes and supplies no prerequisite.